MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

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References

1. Mazidi, 8051 Microcontrollers and Embedded Systems Using Assembly & C, Pearson Education India.
2. Kenneth J. Ayala, Microcontroller, Cengage Learning;
3. kdavpolytech.ac.in/news_files/microprocessor_part_2_compressed_1588258956.pdf
4. https://studytronics.weebly.com/8051internalmemory.html
5. https://www.electronicshub.org/8051-microcontroller-special-function-registers/

Faculty HOD 19/3/2024 Dean Academics

UNIT-4 (Instruction & Architecture of 8051)

Lecture No.-1

8051 Microcontroller Architecture

8051 is an 8 – bit Microcontroller i.e. the data bus of the 8051 Microcontroller (both internal and external) is 8 – bit wide. It is a CISC based Microcontrollerwith Harvard Architecture (separate program and data memory).

Since the basic layout of a microcontroller includes a CPU, ROM, RAM, etc. the 8051 microcontroller also has a similar layout. The following image shows a brief layout of a typical 8051 Microcontroller.

8051 microcontroller is designed by Intel in 1981.

It is an 8-bit microcontroller.

It is built with 40 pins DIP (dual inline package),

4kB of ROM storage

128 bytes of RAM storage

2 16-bit timers.

Four parallel 8-bit ports, which are programmable as well as addressable as perthe requirement.

An on-chip crystal oscillator **is integrated in the microcontroller having crystal** frequency of 12 MHz.

8051 Microcontroller Features

8 – Bit ALU: ALU or Arithmetic Logic Unit is the heart of a microcontroller. It performs arithmetic and bitwise operation on binary numbers. The ALU in 8051 is an 8 – Bit ALU i.e. it can perform operations on 8 – bit data.

8 – Bit Accumulator: The Accumulator is an important register associated with the ALU. The accumulator in 8051 is an 8 – bit register.

RAM: 8051 Microcontroller has 128 Bytes of RAM which includes SFRs and Input / Output Port Registers.

ROM: 8051 has 4 KB of on-chip ROM (Program Memory).

I/O Ports: 8051 has four 8 – bit Input / Output Ports which are bitaddressable and bidirectional.

Timers / Counters: 8051 has two 16 – bit Timers / Counters. Serial

Port: 8051 supports full duplex UART Communication.

External Memory: 8051Microcontroller can access two 16 – bit address line at once: one each for RAM and ROM. The total external memory that an 8051 Microcontroller can access for RAM and ROM is 64KB (216 for each type).

Additional Features: Interrupts, on-chip oscillator, Boolean Processor, etc.

MITRC

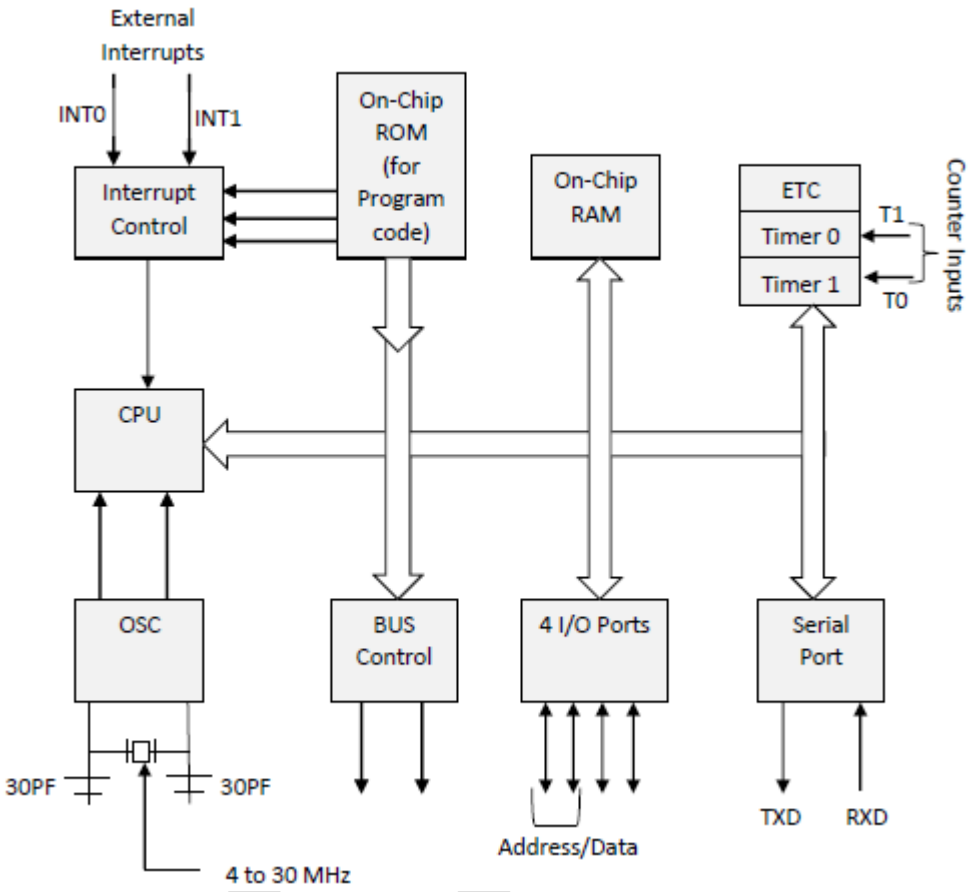


Fig : 8051 Architecture

References

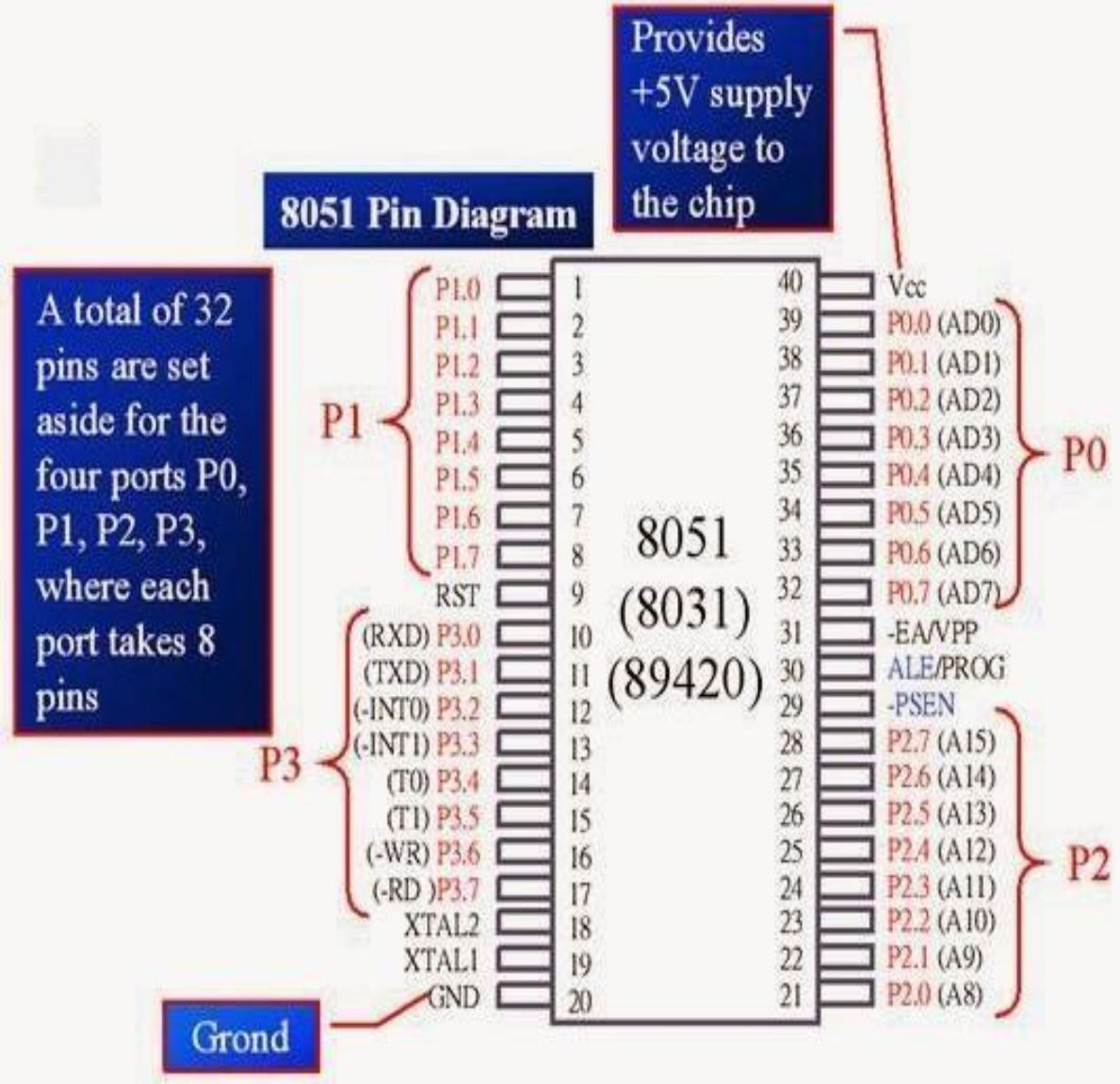
1. Mazidi, 8051 Microcontrollers and Embedded Systems Using Assembly & C, Pearson Education India.

2. Kenneth J. Ayala, Microcontroller, Cengage Learning.

UNIT-4 (Instruction & Architecture of 8051)

Lecture No.-2

PIN Diagram & Functions of 8051Microcontroller



The pin diagram of 8051 microcontroller consists of 40 pins as shown below. A total of 32 pins are set away into four Ports such as P0, P1, P2 and P3. Where, each port contains 8 pins. Therefore, the microcontroller 8051's pin diagram and explanation is given below.

Port1 (Pin1 to Pin8): Port1 includes pin1.0 to pin1.7 and these pins can be configured as input or output pins.

Pin 9 (RST): Reset pin is used to Reset 8051 Microcontroller by giving a positive pulse to this Pin.

Port3 (Pin 10 to 17): The Port3 Pins are similar to port1 pins and can be used as universal Input or output pins. These pins dual-function Pins and the function of each Pin is given as:

Pin 10 (RXD): RXD pin is a Serial Asynchronous Communication Input or Serial synchronous Communication Output.

Pin 11 (TXD): Serial Asynchronous Communication Output or Serial Synchronous Communication clock Output.

Pin 12 (INT0): Input of Interrupt 0

Pin 13 (INT1): Input of Interrupt 1

Pin 14 (T0): Input of Counter 0 clock

Pin 15 (T1): Input of Counter 1 clock

Pin 16 (WR): Writing Signal to write content on external RAM

Pin 17 (RD): Reading Signal to read contents of external RAM.

Pin 18 and 19 (XTAL2, XTAL1): X2 and X1 pins are input output pins for the oscillator. These pins are used to connect an internal oscillator to the microcontroller.

Pin 20 (GND): Pin 20 is a ground pin.

Port2 (Pin 21 to Pin28): Port2 includes pin21 to pin28 which can be configured as Input Output Pins. But, this is only possible when we don't use any external memory. If we use external memory, then these pins will work as high order address bus (A8 to A15).

Pin 29 (PSEN): This pin is used to enable external program memory. If we use an external ROM for storing the program, then logic 0 appears on it, which indicates Micro controller to read data from the memory.

Pin 30 (ALE): Address Latch Enable pin is an active high-output signal. If we use multiple memory chips, then this pin is used to distinguish between them. This Pin also gives program pulse input during programming of EPROM.

Pin 31 (EA): If we have to use multiple memories then the application of logic 1 to this pin instructs the Microcontroller to read data from both memories: first internal and then external.

Port 0 (Pin 32 to 39): Similar to the port 2 and 3 pins, these pins can be used as input output pins when we don't use any external memory. When ALE or Pin 30 is at 1, then this port is used as data bus: when the ALE pin is at 0, then this port is used as a lower order address bus (A0 to A7)

Pin40 (VCC): This VCC pin is used for power supply.

References

1. Mazidi, 8051 Microcontrollers and Embedded Systems Using Assembly & C, Pearson Education India.

2. Kenneth J. Ayala, Microcontroller, Cengage Learning;

UNIT-4 (Instruction & Architecture of 8051)

Lecture No.-3

Different Types of Registers used in 8051 Microcontroller

The register is the main part of the microcontrollers and processors that provide a fast way to collect and store data. If we want to manipulate data with a controller or processor by performing addition, subtraction, and so on, we cannot do that directly in the memory, but it needs registers to process and store the data. Microcontrollers contain several types of registers that can be classified according to their content or instructions that operate in them**.**

Different Types of Registers in the 8051 Microcontroller

A register is a small place in a CPU that can store small amounts of the data used for performing various operations such as addition and multiplication and loads the resulting data on the main memory. Registers contain the address of the memory location where the data is to be stored. The size of the register is very important for modern controllers. For instance, for a 64-bit register, a CPU tries to add two 32-bit numbers and gives a 64-bit result.

Types of Registers

The 8051 microcontroller contains mainly two types of registers:

General-purpose registers (Byte addressable registers)

Special function registers (Bit addressable registers)

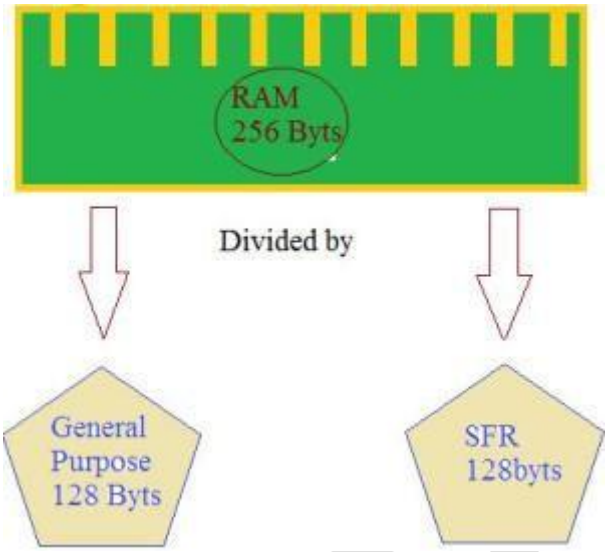
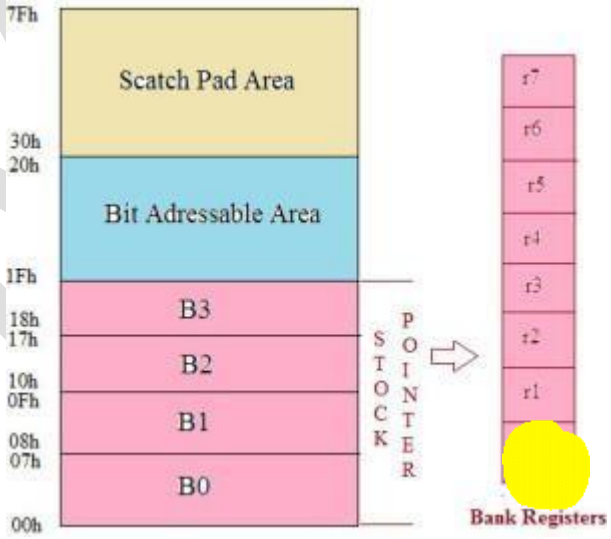


Fig: 8051 Registers

The 8051 microcontroller consists of 256 bytes of RAM, which is divided into two ways, such as 128 bytes for general purpose and 128 bytes for special function registers (SFR) memory. The memory which is used for general purpose is called as RAM, and the memory used for SFR contains all the peripheral related registers like Accumulator, 'B' register, Timers or Counters, and interrupt related registers.

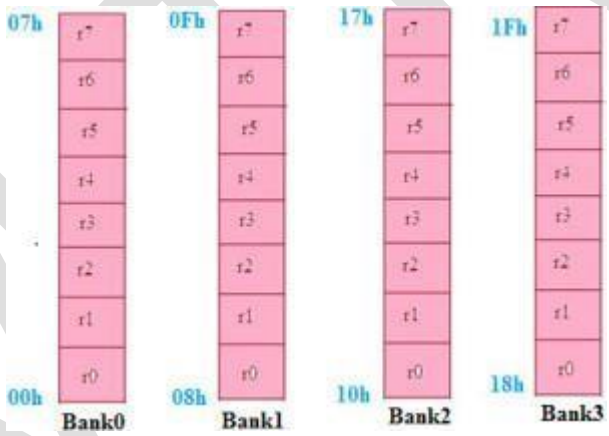
General Purpose Registers



The general-purpose memory is called as the RAM of the 8051 microcontrollers, which is divided into 3 areas such as banks, bit-addressable area, and scratch-pad area. The banks contain different general-purpose registers such as R0-R7, and all such registers are byte-addressable registers that store or remove only 1-byte of data.

Banks and Registers

The B0, B1, B2, and B3 stand for banks, and each bank contains eight general-purpose registers ranging from 'R0' to 'R7'. All these registers are byte-addressable. Data transfer between general-purpose registers to general- purpose registers is not possible. These banks are selected by the Program Status Word (PSW) register.



Special Function Registers in 8051

The 8051 Microcontroller Special Function Registers are used to program and control different hardware peripherals like Timers, Serial Port, I/O Ports etc. In fact, by manipulating the 8051 Microcontroller Special Function Registers (SFRs), you can assess or change the operating mode of the 8051 Microcontroller.

If you observe in Internal RAM Structure, the Address Space from 80H to FFH is allocated to SFRs.

Out of these 128 Memory Locations (80H to FFH), there are only 21 locations that are actually assigned to SFRs. Each SFR has one Byte Address and also a unique name which specifies its purpose.

Since the SFRs are a part of the Internal RAM Structure, you can access SFRs as if you access the Internal RAM. The main difference is the address space: first 128 Bytes (00H to 7FH) is for regular Internal RAM and next 128 Bytes (80H to FFH) is for SFRs.

List of 8051 Microcontroller Special Function Registers

A or ACC

B

DPL

DPH

IE

IP

P0

P1

P2

P3

PCON

PSW

SCON

SBUF

SP

TMOD

TCON

TL0

TH0

TL1

TH1

Categories of 8051 Microcontroller Special Function Registers

All the 21 8051 Microcontroller Special Function Registers (SFRs) along with their functions and Internal RAM Address is given in the following table.

MITRC

Name of the Register	*Function*	*Internal RAM Address (HEX)*
ACC	Accumulator	E0H
B	B Register (for Arithmetic)	F0H
DPH	Addressing External Memory	83H
DPL	Addressing External Memory	82H
IE	Interrupt Enable Control	A8H
IP	Interrupt Priority	B8H
P0	PORT 0 Latch	80H
P1	PORT 1 Latch	90H
P2	PORT 2 Latch	A0H
P3	PORT 3 Latch	B0H
PCON	Power Control	87H
PSW	Program Status Word	D0H
SCON	Serial Port Control	98H
SBUF	Serial Port Data Buffer	99H
SP	Stack Pointer	81H
TMOD	Timer / Counter Mode Control	89H
TCON	Timer / Counter Control	88H
TL0	Timer 0 LOW Byte	8AH
TH0	Timer 0 HIGH Byte	8CH
TL1	Timer 1 LOW Byte	8BH
TH1	Timer 1 HIGH Byte	8DH

The 21 SFR are categorize as follows:

[1] Math or CPU Registers: A and B

[2] Status Register: PSW (Program Status Word)

[3] Pointer Registers: DPTR (Data Pointer – DPL, DPH) and SP (Stack Pointer)

[4] I/O Ports: P0 (Port 0), P1 (Port 1), P2 (Port 2) and P3 (Port 3)

[5] Peripheral Control Registers: PCON, SCON, TCON, TMOD, IE and IP

[6] Peripheral Data Registers: TL0, TH0, TL1, TH1 and SBUF

References

1. Mazidi, 8051 Microcontrollers and Embedded Systems Using Assembly & C, Pearson Education India.

UNIT-4 (Instruction & Architecture of 8051)

Lecture No.-4

Description of Special Function Registers in 8051

[1] Math or CPU Registers: (A and B)

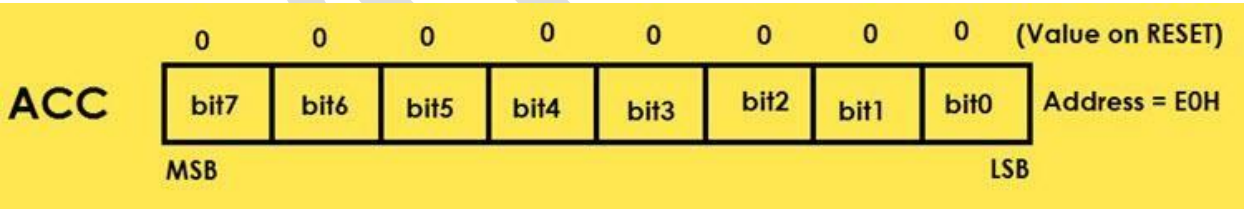
A or Accumulator (ACC)

The Accumulator or Register A is the most important and most used 8051 Microcontroller SFRs. The Register A is located at the address E0H in the SFR memory space. The Accumulator is used to hold the data for almost allthe ALU Operations.

Some of the operations where the Accumulator is used are:

- Arithmetic Operations like Addition, Subtraction, Multiplication etc.
- Logical Operations like AND, OR, NOT etc.
- Data Transfer Operations (between 8051 and External Memory)

The name "Accumulator" came from the fact this register is used to accumulate (or store**)** the result of all Arithmetic and most of the LogicalOperations.

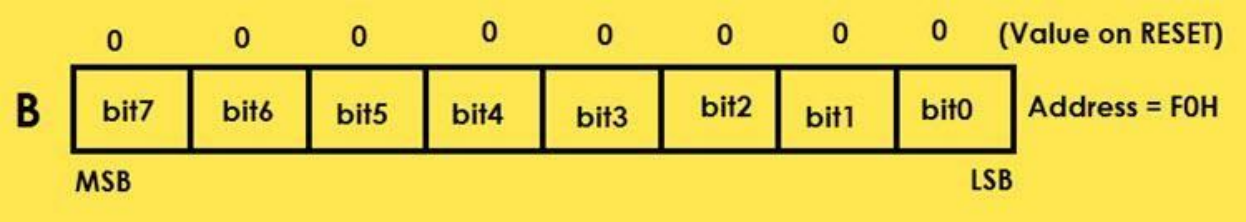


B (Register B)

The B Register is used along with the ACC in Multiplication and Division operations. These two operations are performed on data that are stored only in Registers A and B. During Multiplication Operation, one of the operand (multiplier or multiplicand) is stores in B Register and also the higher byte of the result.

In case of Division Operation, the B Register holds the divisor and also the remainder of the result. It can also be used as a General Purpose Register for normal operations and is often used as an Auxiliary Register by Programmersto store temporary results.

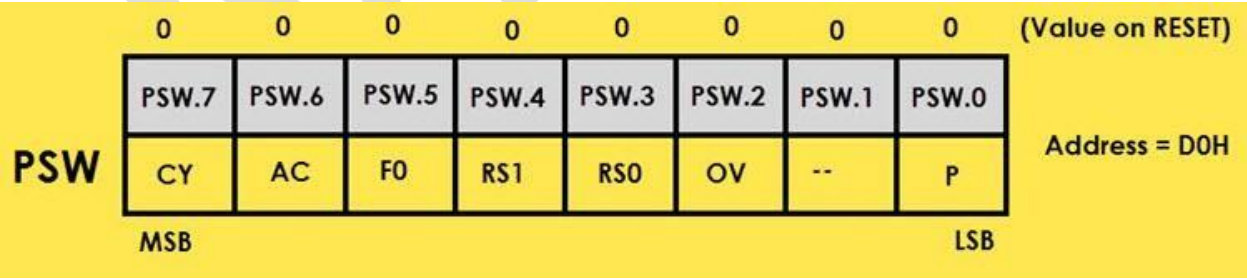
Register B is located at the address F0H of the SFR Address Space.



[2] Program Status Word (PSW) Register

The PSW or Program Status Word Register is also called as Flag Register and is one of the important SFRs. The PSW Register consists of Flag Bits, which help theprogrammer in checking the condition of the result and also make decisions.

Flags are 1-bit storage elements that store and indicate the nature of the result that is generated by execution of certain instructions. The following image shows the contents of the PSW Register.



Given Below table describe the function of each flag

BIT	SYMBOL	FLAG NAME			DESCRIPTION
7	C or CY	Carry			Used in Arithmetic, Logic & Boolean Operations
6	AC	Auxiliary Carry			Used in BCD Arithmetic
5	F0	Flag 0			General Purpose User Flag
4	RS1	Register Bank Selection Bit 1			
3	RS0	Register Bank Selection Bit 1			
		RS1	RS0	Bank	
		0	0	Bank 0	
		0	1	Bank 1	
		1	0	Bank 2	
		1	1	Bank 3	
2	OV	Overflow			Used in Arithmetic Operations
1	--	Reserved			May be used as a General Purpose Flag
0	P	Parity			Set to 1 if A has odd # of 1's; otherwise Reset

Carry Flag (CY): The Address of the Carry flag is D7. This carry flag is affected when the bit is generated from the D7 position.

When No Carry,

CY=0

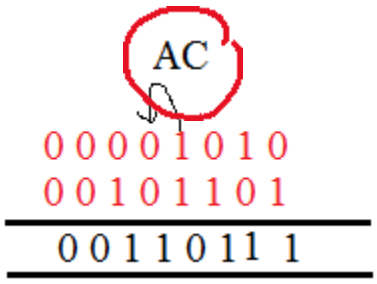
When Carry generated at D7

CY=1

Auxiliary Carry Flag (AC): The address of the auxiliary carry is D6. This auxiliary carryis affected when a bit is generated from the D3 position to the D4 position.

AC=0 , When no carry generated at D3 position

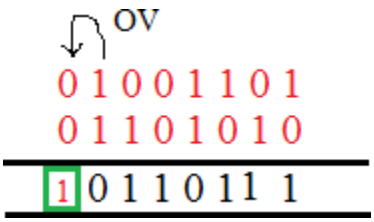
AC=1 When carry generated at D3 position



Overflow Flag (OV): The address of the overflow flag is D2 Position. When a bit isgenerated from the D6 position to the D7 position, then the overflow flag isaffected.

OV=0 overflow flag resets

OV=1 overflow flag sets



Parity Flag (P): The address of the parity flag is D0. While performing arithmetic operations. It is Set (1) when even number of 1 in Accumulator (A), Reset (o) when Odd number of 1 in accumulator.

RS1 and RS0:

RS1	RS0	Bank
0	0	Bank 0
0	1	Bank 1
1	0	Bank 2
1	1	Bank 3

The RS1 and RS0, the bits in the PSW register, are used to select different memory locations (bank0 to bank4) in the RAM.

Description of Special Function Registers in 8051

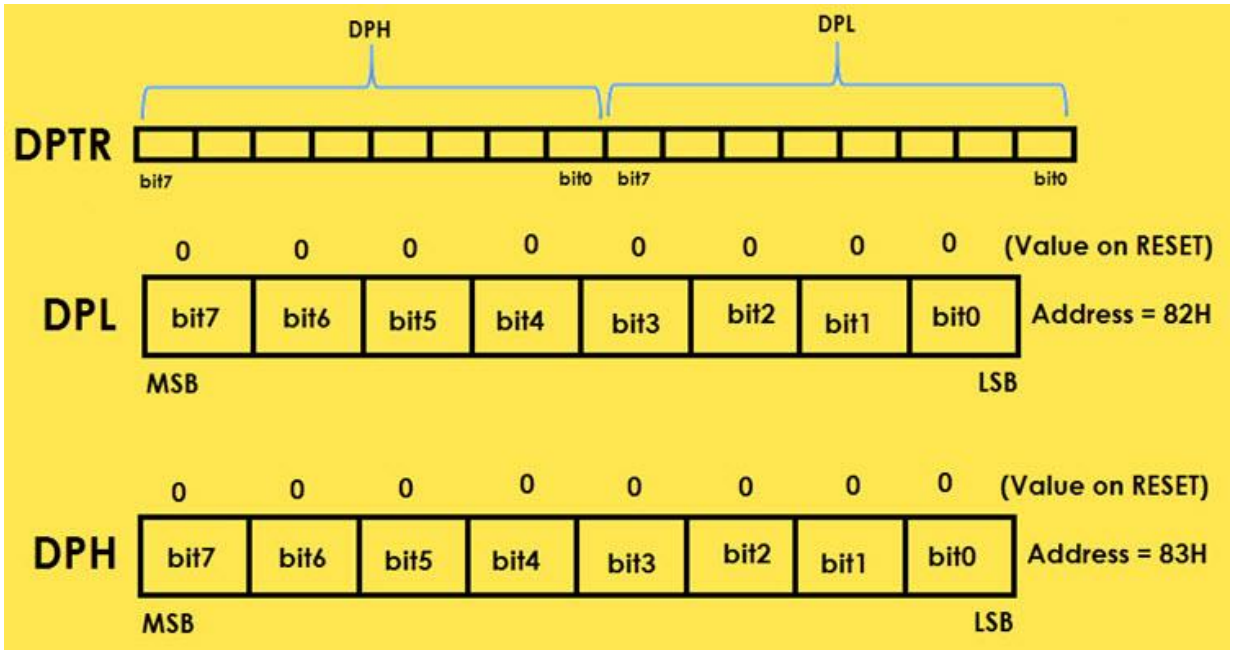
[3] Pointer Registers

(1) Data Pointer (DPTR – DPL and DPH)

The Data Pointer is a 16-bit Register and is physically the combination of DPL (Data Pointer Low) and DPH (Data Pointer High) SFRs. The Data Pointer can be used as a single 16-bit register (as DPTR) or two 8-bit registers (as DPL and DPH).

DPTR doesn't have a physical Memory Address but the DPL (Lower Byte of DPTR) and DPH (Higher Byte of DPTR) have separate addresses in the SFR Memory Space. DPL = 82H and DPH = 83H.

The DPTR Register is used by the programmer addressing external memory (Program – ROM or Data – RAM).



Stack Pointer (SP)

SP or Stack Pointer points out to the top of the Stack and it indicates the next data to be accessed. Stack Pointer can be accesses using PUSH, POP, CALL and RET Instructions. The Stack Pointer is an 8-bit register and upon reset, the Stack Pointer is initialized with 07H.

When writing a new data byte into the stack, the SP (Stack Pointer) is automatically incremented by 1 and the new data is written at an address SP+1. When reading data from stack, the data is retrieved from the Address in SP and after that the SP is decremented by 1 (SP-1).

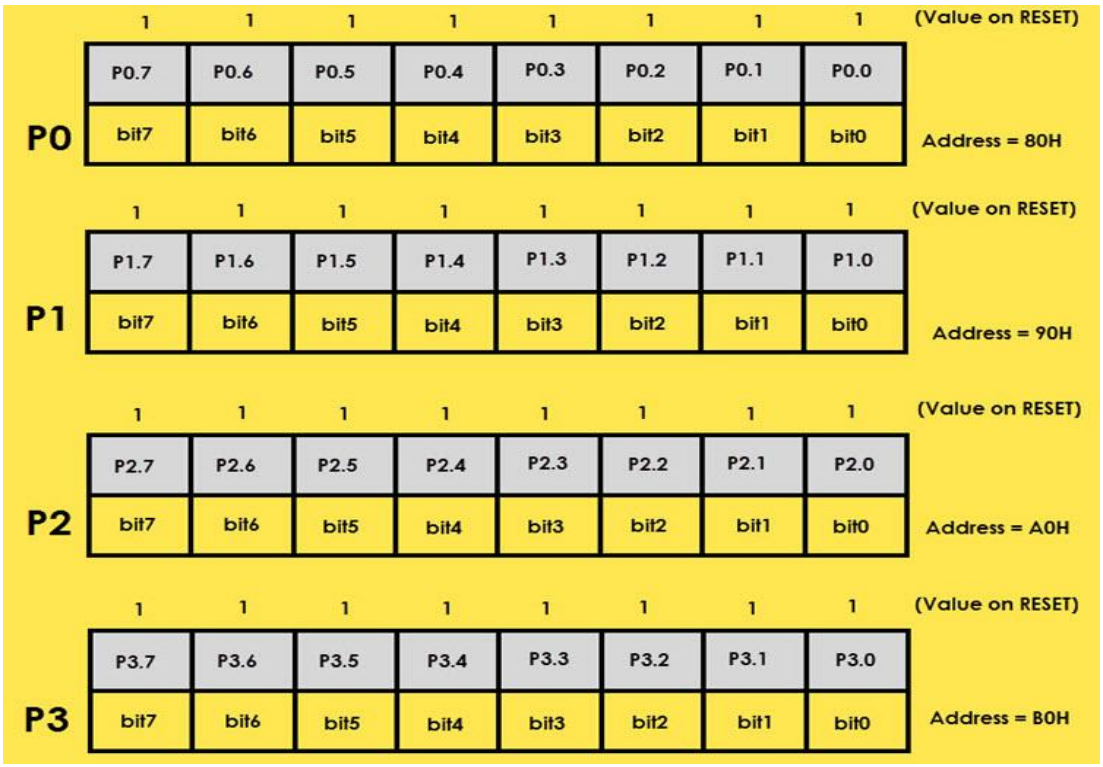


[4] I/O Port Registers (P0, P1, P2 and P3)

The 8051 Microcontroller four Ports which can be used as Input and/or Output. These four ports are P0, P1, P2 and P3. Each Port has a corresponding register with same names (the Port Registers are also P0, P1, P2 and P3). The addresses of the Port Registers are as follows: P0 – 80H, P1 – 90H, P2 – A0H and P2 – B0H.

Each bit in these SFRs corresponds to one physical Pin in the 8051 Microcontroller. All these Port Registers are both Bit Addressable and Byte Addressable. Writing 1 or 0 on a Port Register Bit will reflect as an appropriate voltage (5V and 0V) on the corresponding Pin.

If a Port Bit is SET (declared as 1), the corresponding Port Pin will be configured as Input and similarly if a Port Bit is CLEARED (declared as 0), the corresponding Port Pin is configured as Output. Upon reset, all the Port Bits are SET (1) and hence, all the Port Pins are configured as Inputs.



Description of Special Function Registers in 8051

[5] Peripheral Control Registers

PCON (Power Control)

The PCON or Power Control register, as the name suggests is used to control the 8051 Microcontroller's Power Modes and is located at 87H of the SFR Memory Space. Using two bits in the PCON Register, the microcontroller can be set to Idle Mode and Power Down Mode.

NOTE: PCON register is not bit addressable.

During Idle Mode, the Microcontroller will stop the Clock Signal to the ALU (CPU) but it is given to other peripherals like Timer, Serial, Interrupts, etc. In order to terminate the Idle Mode, you have to use an Interrupt or Hardware Reset.

In the Power Down Mode, the oscillator will be stopped and the power will be reduced to 2V. To terminate the Power Down Mode, you have to use the Hardware Reset.

Apart from these two, the PCON Register can also be used for few additional purposes. The SMOD Bit in the PCON Register is used to control the Baud Rate of the Serial Port.

There are two general purpose Flag Bits in the PCON Register, which can be used by the programmer during execution.



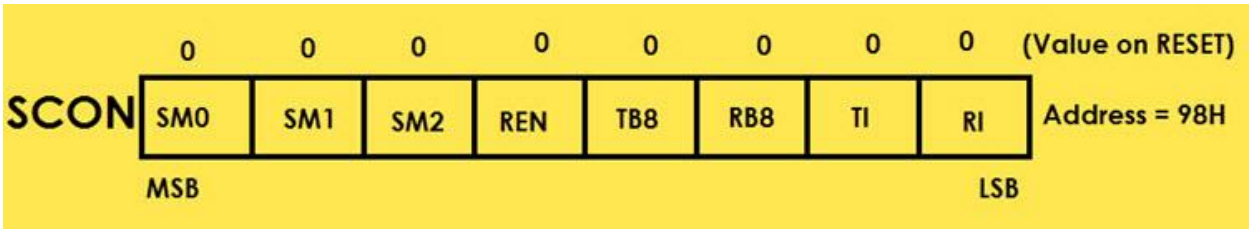
The following table describes the function of each bit in the PCON Register.

Bit	Symbol	Description	Additional Info
7	SMOD	Serial Comm. Baud Rate Modify Bit	If 1, doubles the baud rate using Timer 1. If 0, normal timer 1 baud rate.
6 – 4	—-	—	
3	GF1	General Purpose User Flag (Bit 1)	
2	GF0	General Purpose User Flag (Bit 0)	
1	PD	Power Down Bit	To enter Power Down Mode, set to 1
0	IDL	Idle Mode Bit	To enter Idle Down Mode, set to 1

SCON (Serial Control)

The Serial Control or SCON SFR is used to control the 8051 Microcontroller's Serial Port. It is located as an address of 98H. Using SCON, you can control the Operation Modes of the Serial Port, Baud Rate of the Serial Port and Send or Receive Data using Serial Port.

SCON Register also consists of bits that are automatically SET when a byte of data is transmitted or received.



The following table describes the function of each bit in the SCON Register.

Bit	Symbol	Description
7	SM0	Serial Port Mode Selection Bit 0
6	SM1	Serial Port Mode Selection Bit 1
5	SM2	Multiprocessor Comm. Bit
4	REN	Receive Enable Bit
3	TB8	Transmitted Bit 8
2	RB8	Received Bit 8
1	TI	Transmit Interrupt Flag
0	RI	Receive Interrupt Flag

The Serial Port Mode Selection Bits (SM0 and SM1) determine the mode of UART and also the baud rate. The following table gives an overview of how the Serial Port Mode Selection Bits can be used to configure Serial Port (UART) of 8051.

Serial Port Mode Selection Bits

SM0	SM1	Mode	Description	Baud Rate
0	0	0	8-Bit Synchronous Shift Register Mode	Fixed Baud Rate (Frequency of oscillator / 12)
0	1	1	8-bit Standard UART mode	Variable Baud Rate (Can be set by Timer 1)
1	0	2	9-bit Multiprocessor Comm. mode	Fixed Baud Rate (Frequency of oscillator / 32 or Frequency of oscillator / 64
1	1	3	9-bit Multiprocessor Comm. mode	Variable Baud Rate (Can be set by Timer 1)

TCON (Timer Control)

Timer Control or TCON Register is used to start or stop the Timers of 8051 Microcontroller. It also contains bits to indicate if the Timers has overflowed. The TCON SFR also consists of Interrupt related bits.



TMOD (Timer Mode)

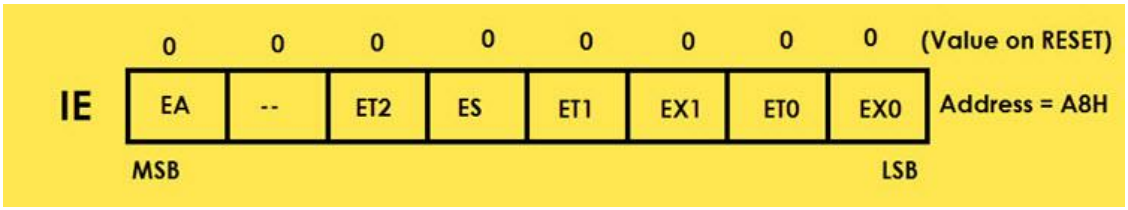
The TMOD or Timer Mode register or SFR is used to set the Operating Modes of the Timers T0 and T1. The lower four



bits are used to configure Timer 0 and the higher four bits are used to configure Timer 1.

IE (Interrupt Enable)

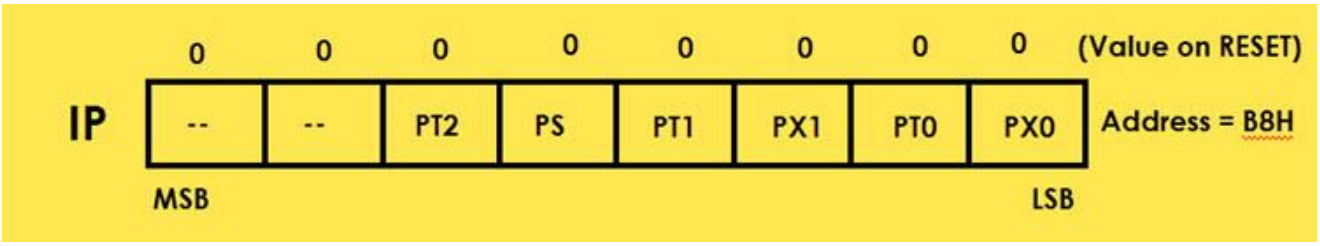
The IE or Interrupt Enable Register is used to enable or disable individual interrupts. If a bit is SET, the corresponding interrupt is enabled and if the bit is cleared, the interrupt is disabled. The Bit7 of the IE register i.e., EA bit is used to enable or disable all the interrupts.



NOTE: **The Interrupt Enable (IE) SFR is bit addressable.**

IP (Interrupt Priority)

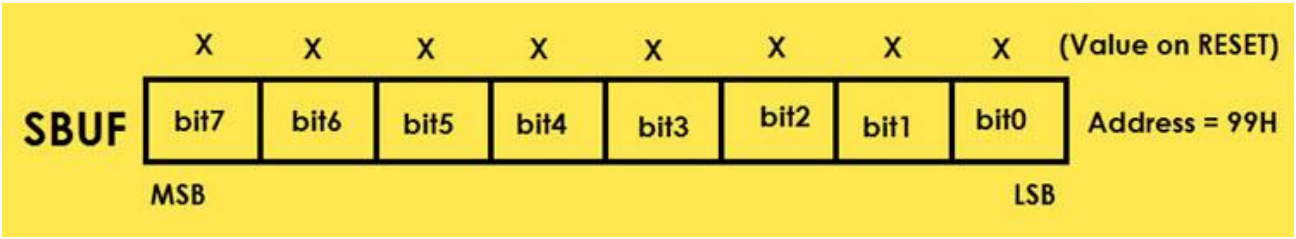
The IP or Interrupt Priority Register is used to set the priority of the interrupt as High or Low. If a bit is CLEARED, the corresponding interrupt is assigned low priority and if the bit is SET, the interrupt is assigned high priority.



[6] Peripheral Data Registers

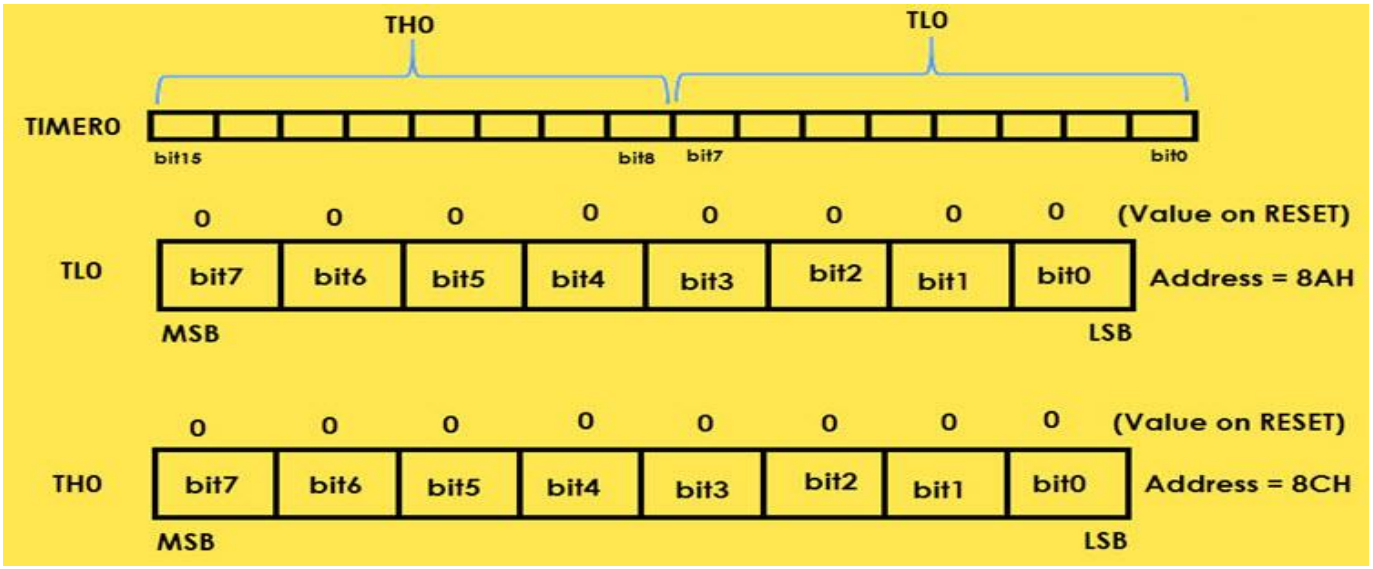
SBUF (Serial Data Buffer)

The Serial Buffer or SBUF register is used to hold the serial data while transmission or reception.



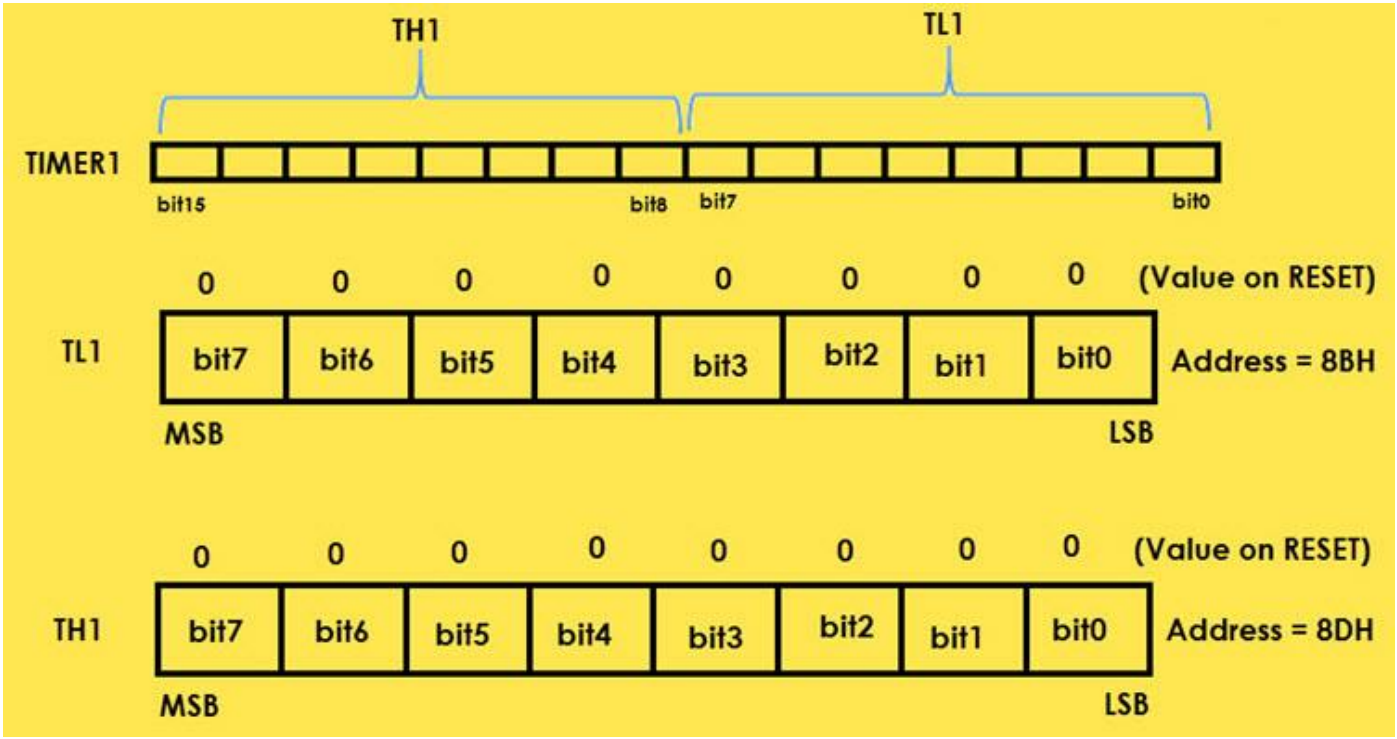
TL0/TH0 (Timer 0 Low/High)

The Timer 0 consists of two SFRs: TL0 and TH0. The TL0 is the lower byte and the TH0 is the higher byte and together they form a 16-bit Timer 0 Register.



TL1/TH1 (Timer 1 Low/High)

The TL1 and TH1 are the lower and higher bytes of the Timer 0.



References

1. Mazidi, 8051 Microcontrollers and Embedded Systems Using Assembly & C, Pearson Education India.

2. Kenneth J. Ayala, Microcontroller, Cengage Learning;

3. https://www.electronicshub.org/8051-microcontroller-special-function-registers/

UNIT-4 (Instruction & Architecture of 8051)

Lecture No.-5

RESET CIRCUIT OF 8051 Microcontroller

RESET is an active High input When RESET is set to High, 8051 goes back tothe power on state. The 8051 is reset by holding the RST high for at least two machine cycles and then returning it low.

There are two method of reset circuit:

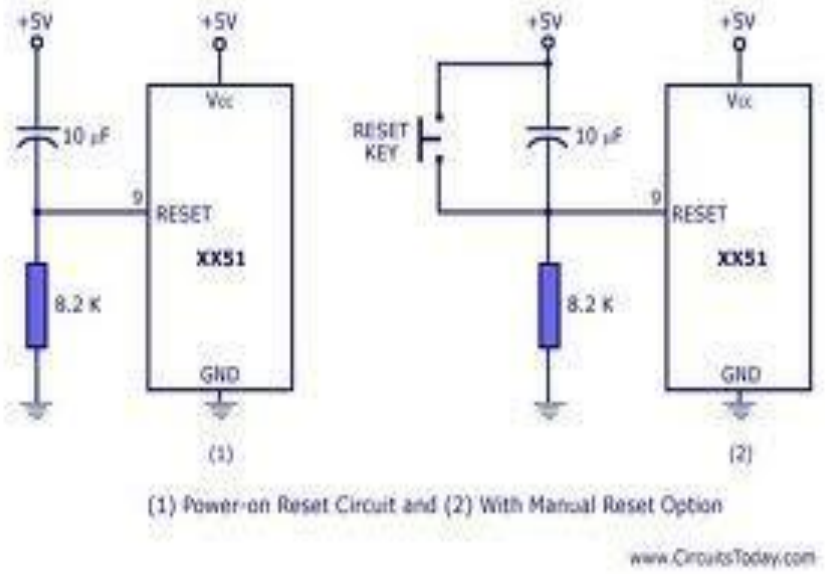
1. **Power On Reset.**

   Initially charging of capacitor makes RST High

   When capacitor charges fully it blocks DC.

2. **Manual Reset**

Closing the switch momentarily will make RST High.



After a reset, the program counter is loaded with 0000H but the content of on-chip RAM is not affected.

Register	Content
Program counter	0000h
Accumulator	00h
B register	00h
PSW	00h
SP	07h
DPTR	0000h
All ports	FFh

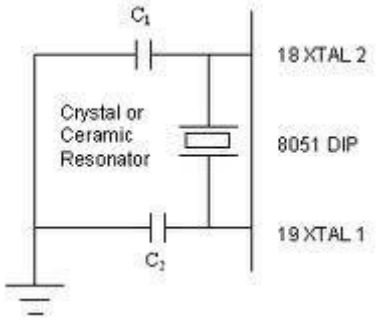
Note: Content of on-chip RAM is not affected by Reset.

OSCILLATOR CIRCUIT OF 8051 CONTROLLER

The 8051 uses the crystal for precisely that to synchronize it's operation. Effectively, the 8051 operates using what are called "machine cycles."

A single machine cycle is the minimum amount of time in which a single 8051 instruction can be executed. although many instructions take multiple cycles.

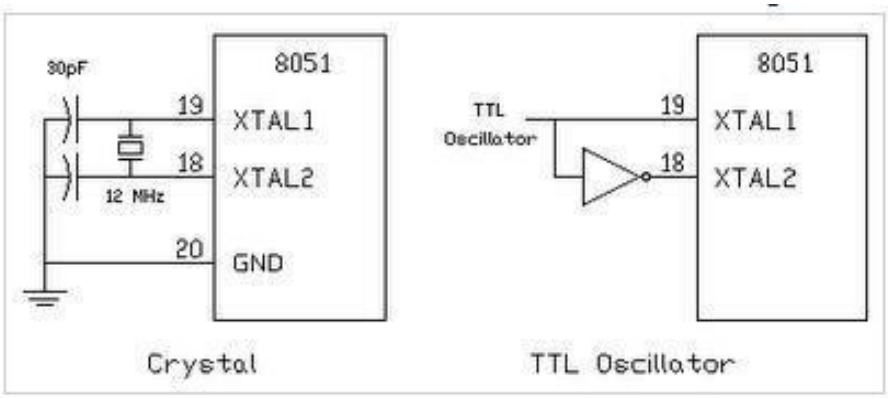
8051 has an on-chip oscillator. It needs an external crystal that decides the operating frequency of the 8051.



This can be achieved in two ways.

The crystal is connected to pins 18 and 19 with stabilizing capacitors. 12 MHz (11.059MHz) crystal is often used and the capacitance ranges from 20pF to 40pF.

The oscillator can also be a TTL clock source connected with a NOT gate as shown.



References

1. Mazidi, 8051 Microcontrollers and Embedded Systems Using Assembly & C, Pearson Education India.

2. Kenneth J. Ayala, Microcontroller, Cengage Learning;

MITRC

UNIT-4 (Instruction & Architecture of 8051)

Lecture No.-6

Stack in the 8051

The stack is a section of a RAM used by the CPU to store information such as data or memory address on temporary basis. The CPU needs this storage area considering limited number of registers.

How Stacks are Accessed

As the stack is a section of a RAM, there are registers inside the CPU to point to it. The register used to access the stack is known as the stack pointer register. The stack pointer in the 8051 is 8-bits wide, and it can take a value of 00 to FFH. When the 8051 is initialized, the SP register contains the value 07H. This means that the RAM location 08 is the first location used for the stack. The storing operation of a CPU register in the stack is known as a **PUSH**, and getting the contents from the stack back into a CPU register is called a **POP**.

Pushing into the Stack

In the 8051, the stack pointer (SP) points to the last used location of the stack. When data is pushed onto the stack, the stack pointer (SP) is incremented by 1. When PUSH is executed, the contents of the register are saved on the stack and SP is incremented by 1. To push the registers onto the stack, we must use their RAM addresses. For example, the instruction "PUSH 1" pushes register R1 onto the stack.

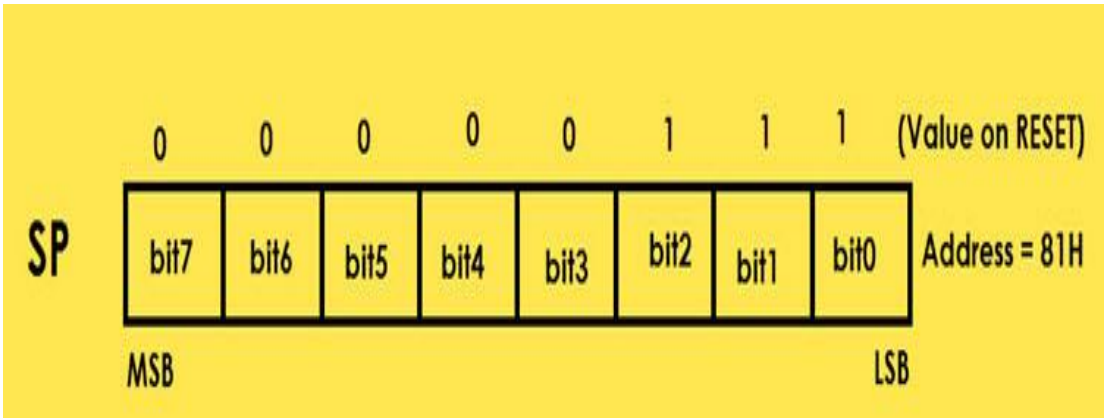
Popping from the Stack

Popping the contents of the stack back into a given register is the opposite to the process of pushing. With every pop operation, the top byte of the stack is copied to the register specified by the instruction and the stack pointer is decremented once.

Stack Pointer (SP)

SP or Stack Pointer points out to the top of the Stack and it indicates the next data to be accessed. Stack Pointer can be accesses using PUSH, POP, CALL and RET Instructions. The Stack Pointer is an 8-bit register and upon reset, the Stack Pointer is initialized with 07H.

When writing a new data byte into the stack, the SP (Stack Pointer) is automatically incremented by 1 and the new data is written at an address SP+1. When reading data from stack, the data is retrieved from the Address in SP and after that the SP is decremented by 1 (SP-1).



Program counter (PC)

The Program Counter (PC) is a 2-byte address which tells the 8051 where the next instruction to execute is found in memory. When the 8051 is initialized PC always starts at 0000h and is incremented each time an instruction is executed. It is important to note that PC is not always incremented by one.

References

1. https://www.refreshnotes.com/2016/02/8051-program-counter.html

2. http://www.faadooengineers.com/online-study/post/eee/embedded-system-design/1066/stack-in-8051

UNIT-4 (Instruction & Architecture of 8051)

Lecture No.-7

Memory Structures in 8051

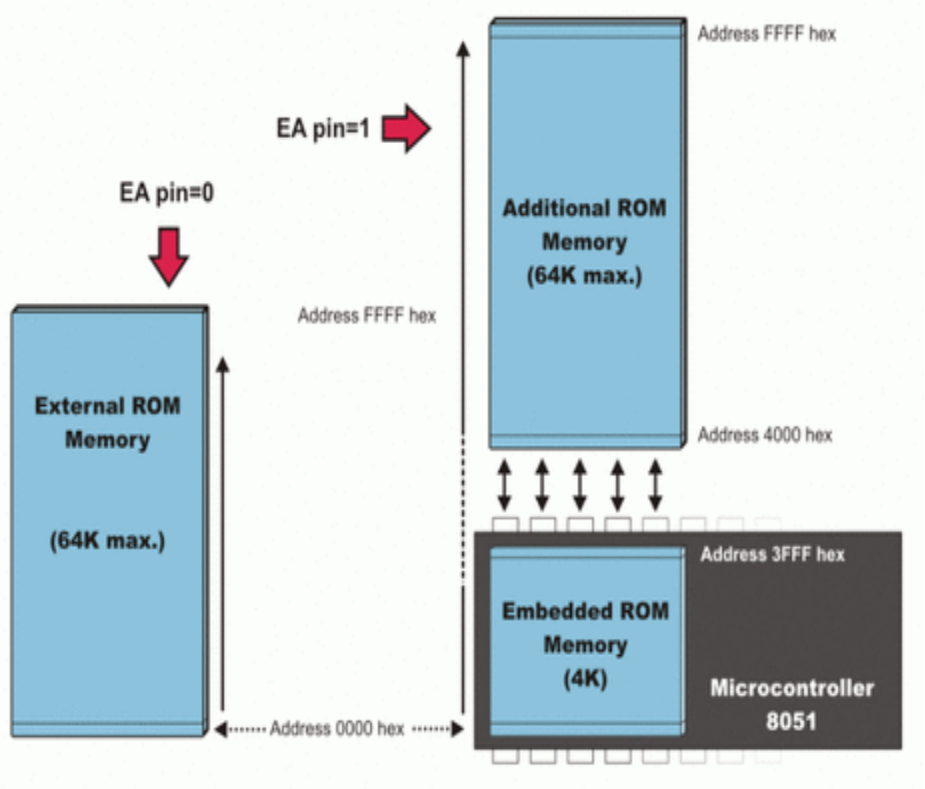
INTERNAL MEMORY OF 8051

The internal memory of 8051 is of 2 types –

1. Internal ROM (Program Memory)
2. Internal RAM (Data Memory)

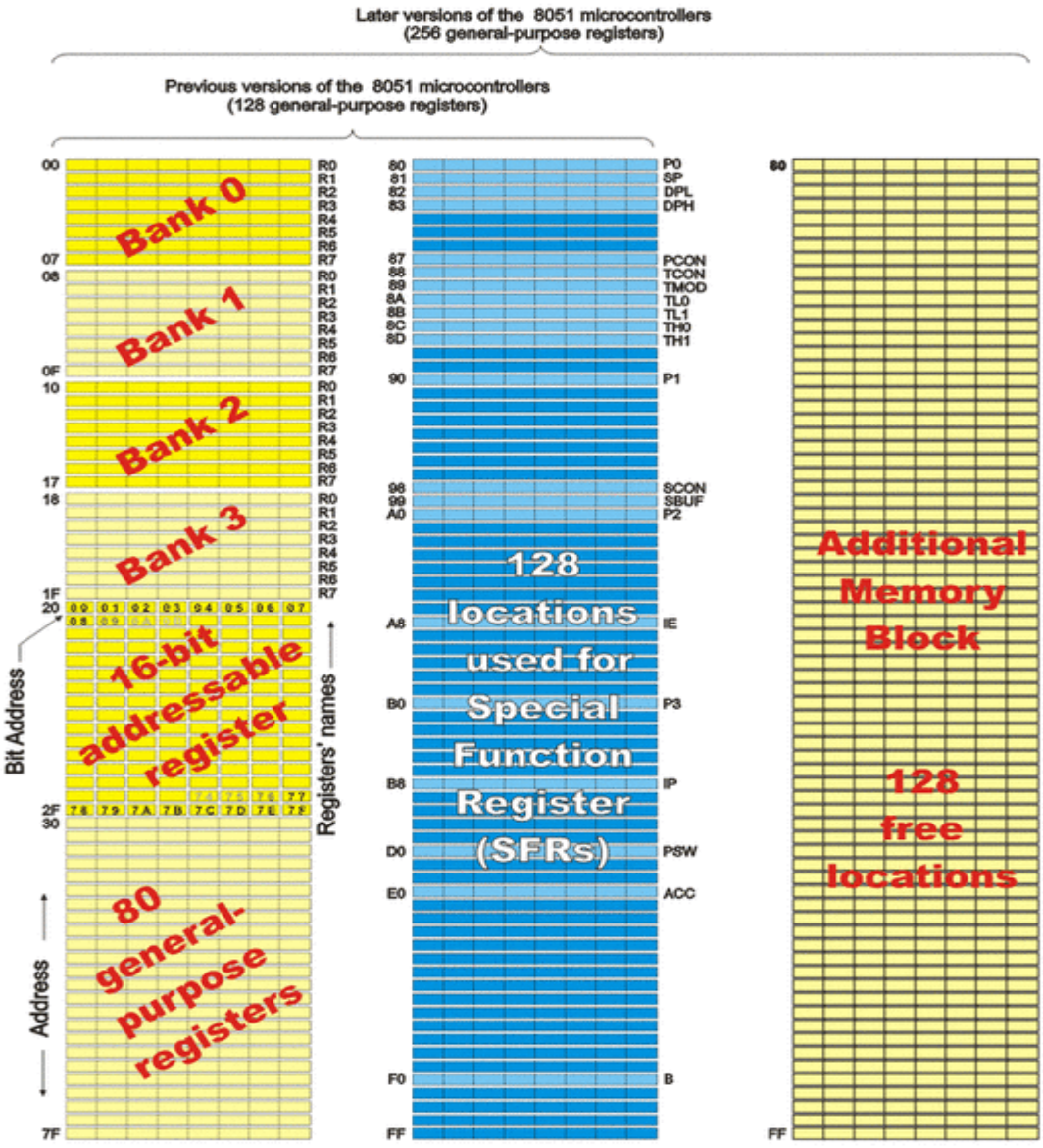
1. Internal ROM (Program Memory)

- 8051 has 4 kB internal Read Only Memory.
- It mainly contains some executable functions and some data which may never change.
- Address range is 0000H to FFFFH.
- EA(bar) is used to allow usage of Internal ROM or not.



2. Internal RAM (Data Memory)

- 8051 has 128 B internal Random Access Memories.
- Data Memory is used for temporarily storing data and intermediate results created and used during the operation of the microcontroller.
- It mainly contains register banks, Read/write data, bit addressable registers and stack.
- Address range is 0000H to FFFFH.
- PSEN (bar) is used to allow usage of Internal ROM or not.



Reference

1. https://studytronics.weebly.com/8051internalmemory.html

UNIT-4 (Instruction & Architecture of 8051)

Lecture No.-8

Timing Diagram & Execution Cycle

8051 series microcontroller has 111 instructions they are:

49 single byte instructions (Only Opcode)

45 two bytes instructions (Opcode + One Byte Operand)

17 three bytes instructions (Opcode + Two Byte Operands)

Based on the operation they perform, all the instructions in the 8051 Microcontroller Instruction Set are divided into five groups. They are:

- Data Transfer Instructions
- Arithmetic Instructions
- Logical Instructions
- Boolean or Bit Manipulation Instructions
- Program Branching Instructions

[1] Data Transfer Instructions

The Data Transfer Instructions are associated with transfer of data between registers or external program memory or external data memory. The Mnemonics associated with Data Transfer are given below.

- MOV (Move Data)
- MOVC (Move Code)
- MOVX (Move External Data)
- PUSH (Move Data to Stack)
- POP (Copy Data from Stack)
- XCH (Exchange Data between two Registers)
- XCHD (Exchange Lower Order Data between two Registers)

Arithmetic Instructions

Using Arithmetic Instructions, you can perform addition, subtraction, multiplication and division. The arithmetic instructions also include increment by one, decrements by one and a special instruction called Decimal Adjust Accumulator. The Mnemonics associated with the Arithmetic Instructions are:

- ADD (Addition without Carry)
- ADDC (Addition with Carry)
- SUBB (Subtract with Carry)
- INC (Increment by 1)
- DEC (Decrement by 1)
- MUL (Multiply)
- DIV (Divide)
- DA A (Decimal Adjust the Accumulator (A Register))

Mnemonic	Instruction	Description	Addressing Mode	# of Bytes	# of Cycles
MOV	A, #Data	A ← Data	Immediate	2	1
	A, Rn	A ← Rn	Register	1	1
	A, Direct	A ← (Direct)	Direct	2	1
	A, @Ri	A ← @Ri	Indirect	1	1
	Rn, #Data	Rn ← data	Immediate	2	1
	Rn, A	Rn ← A	Register	1	1
	Rn, Direct	Rn ← (Direct)	Direct	2	2
	Direct, A	(Direct) ← A	Direct	2	1
	Direct, Rn	(Direct) ← Rn	Direct	2	2
	Direct1, Direct2	(Direct1) ← (Direct2)	Direct	3	2
	Direct, @Ri	(Direct) ← @Ri	Indirect	2	2
	Direct, #Data	(Direct) ← #Data	Direct	3	2
	@Ri, A	@Ri ← A	Indirect	1	1
	@Ri, Direct	@Ri ← Direct	Indirect	2	2
	@Ri, #Data	@Ri ← #Data	Indirect	2	1
	DPTR, #Data16	DPTR ← #Data16	Immediate	3	2
MOVC	A, @A+DPTR	A ← Code Pointed by A+DPTR	Indexed	1	2
	A, @A+PC	A ← Code Pointed by A+PC	Indexed	1	2
	A, @Ri	A ← Code Pointed by Ri (8-bit Address)	Indirect	1	2
MOVX	A, @DPTR	A ← External Data Pointed by DPTR	Indirect	1	2
	@Ri, A	@Ri ← A (External Data 8-bit Addr)	Indirect	1	2
	@DPTR, A	@DPTR ← A (External Data 16-bit Addr)	Indirect	1	2
PUSH	Direct	Stack Pointer SP ← (Direct)	Direct	2	2
POP	Direct	(Direct) ← Stack Pointer SP	Direct	2	2
XCH	Rn	Exchange ACC with Rn	Register	1	1
	Direct	Exchange ACC with Direct Byte	Direct	2	1
	@Ri	Exchange ACC with Indirect RAM	Indirect	1	1
XCHD	A, @Ri	Exchange ACC with Lower Order Indirect RAM	Indirect	1	1

Fig: Data Transfer Instructions

ARITHMETIC OPERATORS

Instruction	Description	Bytes	Periods	C	OV	AC
ADD A, Rn	Add register to ACC	1	12	x	x	x
ADD A, direct	Add direct byte to ACC	2	12	x	x	x
ADD A, @Ri	Add indirect RAM to ACC	1	12	x	x	x
ADD A, #data	Add immediate data to ACC	2	12	x	x	x
ADDC A, Rn	Add register to ACC with Carry	1	12	x	x	x
ADDC A, direct	Add direct byte to ACC with Carry	2	12	x	x	x
ADDC A, @Ri	Add indirect RAM to ACC with Carry	1	12	x	x	x
ADDC A, #data	Add immediate data to ACC with Carry	2	12	x	x	x
SUBB A, Rn	Subtract Register from ACC with borrow	1	12	x	x	x
SUBB A, direct	Subtract indirect RAM from ACC with borrow	2	12	x	x	x
SUBB A, @Ri	Subtract indirect RAM from ACC with borrow	1	12	x	x	x
SUBB A, #data	Subtract immediate data from ACC with borrow	2	12	x	x	x
INC A	Increment ACC	1	12			
INC Rn	Increment register	1	12			
INC direct	Increment direct byte	2	12			
INC @Ri	Increment direct RAM	1	12			
DEC A	Decrement ACC	1	12			
DEC Rn	Decrement Register	1	12			
DEC direct	Decrement direct byte	2	12			
DEC @Ri	Decrement indirect RAM	1	12			
INC DPTR	Increment Data Pointer	1	24			
MUL AB	Multiply A & B	1	48	0	x	
DIV AB	Divide A by B	1	48	0	x	
DA A	Decimal Adjust ACC	1	12	x		

Fig 2 Arithmetic Instructions

Instruction Cycle

The 8051 microcontroller executes instructions through a process involving fetching, decoding, and executing instructions, collectively known as the instruction cycle.

Timing Diagram-It is a graphical representation. It represents the execution time taken by each instruction in a graphical format. The execution time is represented in T-states

Machine Cycle- Time required to access the memory or input-output devices is called machine cycle.

References:

1. https://www.studocu.com/in/document/pes-university/microprocessor-and- microcontroller/timing-diagram-prof-johnson/18319791

2. kdavpolytech.ac.in/news_files/microprocessor_part_2_compressed_1588258956.pdf